

A fixed-class Möbius-weighted estimate and a residual relation in the Huang–Li Goldbach reduction

Youngrok Lee

Independent researcher, Republic of Korea

Abstract

Huang and Li [2] obtain a conditional binary Goldbach conclusion for sufficiently large even integers from the Elliott–Halberstam conjecture together with a Möbius-twisted variant. In their Section 3 the Möbius-twisted hypothesis is used to control two signed fixed-class terms, $E_4(\alpha)$ and $E_3(\alpha)$. In the regime corresponding to their corollary, we prove that the fixed-class sum underlying E_4 is $\ll_A N(\log N)^{-A}$ for every $A > 0$, uniformly in the truncation point, using divisor switching and Bombieri–Vinogradov. Thus that use of the Möbius-twisted hypothesis can be replaced by an unconditional estimate. The proof completes the fixed divisor sum and then works on the complementary range, where the surviving Möbius factor lies on a cofactor within the Bombieri–Vinogradov range. This mechanism concerns the signed fixed-class combination only and does not establish the full EH_μ hypothesis. For the remaining logarithmically weighted term we derive

$$E_3(\alpha) = \tilde{r}(N) - \mathfrak{S}(N) \left(N - \sum_{n < N} \Lambda(n) \mu(N - n) \right) + O_A(N(\log N)^{-A}).$$

Consequently, at this error scale, the E_3 estimate used in the Huang–Li decomposition considered here is equivalent to their equation (22); this is a relation inside that reduction, not a new estimate for the remaining correlation. We also record the moving-cut correction to their published equation (18), following the prior defect report and Moya’s repair [7], and bound the corresponding outside correction term. No unconditional progress toward Goldbach is claimed.

1 Introduction

Let N be a large even integer, Λ the von Mangoldt function and μ the Möbius function. Put

$$\tilde{r}(N) = \sum_{n < N} \Lambda(n) \Lambda(N - n), \quad \mathfrak{S}(N) = 2 \prod_{p > 2} \left(1 - \frac{1}{(p-1)^2} \right) \prod_{\substack{p|N \\ p > 2}} \left(1 + \frac{1}{p-2} \right).$$

Write $EH_\mu(Q)$ for the assertion that, for every $A > 0$,

$$\sum_{q \leq Q} \max_{y < N} \max_{\substack{(a,q)=1 \\ n \leq y \\ n \equiv a \pmod{q}}} \left| \sum_{\substack{n \leq y \\ n \equiv a \pmod{q}}} \Lambda(n) \mu(N - n) - \frac{1}{\varphi(q)} \sum_{n \leq y} \Lambda(n) \mu(N - n) \right| \ll_A \frac{N}{(\log N)^A}. \quad (1)$$

Huang and Li [2] state, under their Elliott–Halberstam hypotheses, the lower bound

$$\tilde{r}(N) \geq \mathfrak{S}(N)(1 - \mathfrak{A}(N))N + O\left(\frac{N}{(\log N)^A}\right), \quad (2)$$

where

$$\mathfrak{A}(N) = \prod_{\substack{q \nmid N \\ q > 2}} \left(1 - \frac{1}{q(q-1)} \right). \quad (3)$$

Their corollary uses Bombieri–Vinogradov for the ordinary prime progression input and leaves $EH_\mu(N^{\theta'})$ with one fixed $\theta' > 1/2$.

The Möbius-twisted hypothesis is used in their Section 3 to bound two fixed-class error terms. With $K = (N - 1)/\alpha$ they are

$$E_3(\alpha) = \sum_{\substack{k < K \\ (k, N) = 1}} \mu(k) \log k \left[\sum_{\substack{n < N \\ n \equiv N \pmod{k}}} \Lambda(n) \mu(N - n) - \frac{1}{\varphi(k)} \sum_{n < N} \Lambda(n) \mu(N - n) \right], \quad (4)$$

$$E_4(\alpha) = \sum_{\substack{k < K \\ (k, N) = 1}} \mu(k) \left[\sum_{\substack{n < N \\ n \equiv N \pmod{k}}} \Lambda(n) \mu(N - n) \log(N - n) - \frac{1}{\varphi(k)} \sum_{n < N} \Lambda(n) \mu(N - n) \log(N - n) \right]. \quad (5)$$

For $(k, N) = 1$ and $1 \leq t < N$ define

$$E_\mu(t; k) = \sum_{\substack{n \leq t \\ n \equiv N \pmod{k}}} \Lambda(n) \mu(N - n) - \frac{1}{\varphi(k)} \sum_{n \leq t} \Lambda(n) \mu(N - n), \quad C(t) = \sum_{n \leq t} \Lambda(n) \mu(N - n), \quad (6)$$

and

$$T_w(t) = \sum_{\substack{k < K \\ (k, N) = 1}} \mu(k) w(k) E_\mu(t; k).$$

For endpoint notation set $C(N) := C(N - 1)$ and $E_\mu(N; k) := E_\mu(N - 1; k)$. Thus $E_3 = T_{\log}(N - 1)$, while E_4 is obtained from $T_1(t)$ by Abel summation. We use the *Corollary-1 regime* $\alpha = N^{1-\theta'}$, $\theta' \in (1/2, 1)$. Then $K = (N - 1)N^{\theta'-1} = N^{\theta'} + O(1)$ and $N/K \asymp N^{1-\theta'}$.

Theorem 1. Fix $\theta' \in (1/2, 1)$. Uniformly for real K satisfying $|K - N^{\theta'}| \leq 1$ and every $A > 0$,

$$\sup_{1 \leq t < N} \left| \sum_{\substack{k < K \\ (k, N) = 1}} \mu(k) E_\mu(t; k) \right| \ll_{A, \theta'} \frac{N}{(\log N)^A}.$$

All terms in the proof other than the Bombieri–Vinogradov contribution are $O(Ne^{-c\sqrt{\log N}})$.

The theorem concerns a signed sum in the fixed class $a = N$; it is not an instance of EH_μ , which places absolute values outside each modulus and maximises over the residue class.

Corollary 2. In the Corollary-1 regime, $E_4(\alpha) \ll_A N(\log N)^{-A}$ for every $A > 0$ without assuming EH_μ . Hence the use of Huang–Li’s Lemma 5 to control E_4 can be replaced by Theorem 1. After the moving-cut issue of Section 7 is accounted for, the remaining use of EH_μ within the E_3/E_4 decomposition considered here is the estimate for $E_3(\alpha)$.

Theorem 3. In the Corollary-1 regime, for every $A > 0$,

$$E_3(\alpha) = \tilde{r}(N) - \mathfrak{S}(N)(N - C(N)) + O_A\left(\frac{N}{(\log N)^A}\right). \quad (7)$$

Corollary 4 (conditional consequences). Assume in addition that

$$E_3(\alpha) \ll_A N(\log N)^{-A}$$

for every $A > 0$. This is the remaining input in the Huang–Li E_3/E_4 decomposition and is not proved here. Then (7) is, at the same error scale, equivalent to their equation (22). Together with the unconditional Bombieri–Vinogradov estimate used by Huang–Li,

$$|C(N)| \leq U(N), \quad U(N) = \mathfrak{A}(N)N + O_B(N(\log N)^{-B}),$$

it yields binary Goldbach for all sufficiently large even N . Under the same additional estimate, $\tilde{r}(N) \sim \mathfrak{S}(N)N$ if and only if $C(N) = o(N)$.

Theorem 3 is specific to the logarithmic weight occurring in this reduction: completing the divisor sum invokes $\mu * \log = \Lambda$ and reproduces $\tilde{r}(N)$. Corollary 4 is only a conditional consequence of the additional E_3 estimate; that estimate is not supplied by Theorem 3. We do not claim that every alternative smoothing or reformulation is covered by this calculation.

Proposition 5 (one-sided sufficient condition). *In the Corollary-1 regime, binary Goldbach for all sufficiently large even N follows if, for one fixed $\varepsilon > 0$,*

$$E_3(\alpha) > -(1 - \varepsilon)\mathfrak{S}(N)(1 - \mathfrak{A}(N))N. \quad (8)$$

Proposition 6 (absolute-value sufficient condition). *Put*

$$B_\mu(N) = \sum_{\substack{k < K \\ (k, N)=1}} \mu^2(k)(\log k)|E_\mu(N; k)|. \quad (9)$$

Then binary Goldbach for all sufficiently large even N follows if, for one fixed $\varepsilon > 0$,

$$B_\mu(N) \leq (1 - \varepsilon)\mathfrak{S}(N)(1 - \mathfrak{A}(N))N. \quad (10)$$

The full hypothesis $EH_\mu(N^{\theta'})$ implies this condition, but the converse is not asserted.

The scope is deliberately limited. Theorem 1 gives no new bound for $C(N)$, and Theorem 3 converts the surviving E_3 estimate into a relation containing the Goldbach sum rather than proving that estimate. Accordingly no unconditional progress toward the Goldbach conjecture is claimed. Section 7 concerns a published rearrangement in [2]; it provides a correction and does not claim that their stated conditional theorem is false.

2 Notation and the mechanism

p and q always denote primes. φ is Euler’s function, τ_3 the 3-fold divisor function, $\text{rad}(u) = \prod_{p|u} p$. Constants $c, c_1, \dots$ are positive and absolute unless subscripted; implied constants may depend on A and θ' . The letter A is reserved for the free exponent of (1); the local factor of (3) is $\mathfrak{A}$.

The mechanism of Theorem 1 is the following. The residue class in (6) is the fixed class $n \equiv N \pmod{k}$, i.e.

$$n \equiv N \pmod{k} \iff k \mid N - n.$$

So the k -sum is a divisor sum over $u = N - n$, namely the incomplete sum $\sigma_K(u) = \sum_{k|u, k < K, (k, N)=1} \mu(k)$. Put $\delta = (\theta' - \frac{1}{2})/2 > 0$; then $N/K \ll N^{1/2-2\delta}$.

First one completes it. By Lemma 7, for squarefree u the complete sum $\sum_{k|u, (k, N)=1} \mu(k)$ equals $\mathbf{1}_{\text{rad}(u)|N}$, which forces $u \mid \text{rad}(N)$ and so leaves only $N^{o(1)}$ terms. *Then the work sits entirely in the complementary sum, over $k \geq K$ — and it is only on that range that writing $u = mk$ gives*

$$m = u/k < N/K \ll N^{1/2-2\delta}.$$

Finally, for squarefree $u = mk$ the factorisation forces $(m, k) = 1$ and $\mu(u)\mu(k) = \mu(m)\mu^2(k)$: the Möbius factor on the *long* variable k cancels against itself and leaves the nonnegative μ^2 , while the surviving Möbius sits on the *short* variable m . That is the classical “Möbius on the short variable plus Bombieri–Vinogradov” configuration.

The completion step is essential for this proof. Before completion the cofactor u/k need not be $O(N^{1-\theta'})$, so the surviving Möbius factor is not placed on the range controlled by the Bombieri–Vinogradov step below.

For the present theorem, the only size fact needed here is $m \ll N^{1/2-2\delta}$, which places the cofactor inside the Bombieri–Vinogradov range used below. Steps 1–3 of Section 3 carry out the argument in that order.

3 Proof of Theorem 1

Fix $A > 0$ and set

$$D_0 = E_0 := \exp(\sqrt{\log N}). \quad (11)$$

The exponential truncation is independent of A and puts every non-Bombieri–Vinogradov tail below every power of $\log N$, while remaining harmless for the modulus bound (21).

3.1 Step 0: the subtracted mean term

By (6),

$$T_1(t) = D(t) - C(t)B(K), \quad D(t) := \sum_{\substack{k < K \\ (k, N)=1}} \mu(k) \sum_{\substack{n \leq t \\ n \equiv N \pmod{k}}} \Lambda(n) \mu(N - n), \quad (12)$$

where $B(K) = \sum_{k < K, (k, N)=1} \mu(k)/\varphi(k)$. Huang–Li’s Lemma 1 (a form of Goldston–Yıldırım [1]) gives the same bound with $k \leq K$; if K is an integer the endpoint difference is $O(K^{-1+o(1)})$ and is absorbed. Hence $B(K) \ll e^{-c_1 \sqrt{\log K}}$, and trivially $|C(t)| \leq \sum_{n < N} \Lambda(n) \ll N$. Hence

$$|C(t)B(K)| \ll Ne^{-c\sqrt{\log N}} \quad (13)$$

uniformly in t .

3.2 Step 1: divisor switching (an exact identity)

Since $n \equiv N \pmod{k}$ with $n \leq t < N$ is the same as $u := N - n$ being a multiple of k in $[N - t, N)$,

$$D(t) = \sum_{N-t \leq u < N} \Lambda(N - u) \mu(u) \sigma_K(u), \quad \sigma_K(u) := \sum_{\substack{k|u, k < K \\ (k, N)=1}} \mu(k). \quad (14)$$

This is a finite rearrangement and hence exact.

3.3 Step 2: the complete divisor sum

Lemma 7. *For every $u \geq 1$, $\sum_{k|u, (k, N)=1} \mu(k) = \mathbf{1}_{\text{rad}(u)|N}$.*

Proof. The sum is multiplicative in u with local factor 1 at $p \mid N$ and $1 - 1 = 0$ at $p \nmid N$. Hence it vanishes unless every prime of u divides N . $\square$

Splitting $\sigma_K(u)$ into the complete sum minus the tail $k \geq K$, (14) becomes $D(t) = P(t) - R(t)$ with

$$P(t) = \sum_{\substack{N-t \leq u < N \\ \text{rad}(u) | N}} \Lambda(N-u) \mu(u), \quad R(t) = \sum_{N-t \leq u < N} \Lambda(N-u) \mu(u) \sum_{\substack{k|u, k \geq K \\ (k,N)=1}} \mu(k). \quad (15)$$

In $P(t)$ the conditions $\mu(u) \neq 0$ and $\text{rad}(u) | N$ force $u | \text{rad}(N)$, so there are at most $2^{\omega(N)} = N^{o(1)}$ terms, each $\ll \log N$:

$$P(t) \ll N^{o(1)}. \quad (16)$$

3.4 Step 3: the residual, and the degeneracy lemma

Write $u = mk$ with $k \geq K$, so $m = u/k < N/K =: M$. For squarefree u the factorisation forces $(m, k) = 1$ and $\mu(u)\mu(k) = \mu(m)\mu^2(k)$, so

$$R(t) = \sum_{m < M} \mu(m) \sum_{\substack{k \geq K, (k,m)=1, (k,N)=1 \\ N-t \leq mk < N}} \mu^2(k) \Lambda(N-mk). \quad (17)$$

Lemma 8 (degeneracy). *The contribution to (17) of the terms with $(k, N) > 1$ is $O(N^{o(1)})$. The same bound holds for the terms in which the modulus constructed in Step 4 is not coprime to N .*

Proof. Suppose $p | (k, N)$ and $\Lambda(N-mk) \neq 0$, say $N-mk = q^\ell$. Since $p | k$ and $p | N$ we get $p | N-mk$, hence $q = p$ and $n := N-mk = p^\ell$ with $p | N$. The number of such $n < N$ is $\leq \sum_{p|N} \log N / \log p \ll (\log N)^2$; for each, the pairs (m, k) with $mk = N-n$ number $\leq \tau(N-n) \ll N^{o(1)}$, and each term is $\ll \log N$.

For the second sentence, let $q = m \text{lcm}(d^2, e)$ be the modulus built in Step 4 and suppose $g = (q, N) > 1$. A term of (20) in that class has $n \equiv N \pmod{q}$ with $1 \leq n \leq \mathcal{T}_m(t)$, so $g | N-n$; together with $g | N$ this gives $g | n$, and any $p | g$ then divides n , so $\Lambda(n) \neq 0$ forces $n = p^\ell$ with $p | N$. The number of such $n < N$ is again $\leq \sum_{p|N} \log N / \log p \ll (\log N)^2$. For each such n , a triple (m, d, e) places it in a degenerate class only if $m \text{lcm}(d^2, e) | N-n$, so each of m, d^2 and e divides $N-n$ and the triples number $\ll \tau(N-n)^3 \ll N^{o(1)}$; with $\Lambda(n) \ll \log N$ the total is $\ll N^{o(1)}$. $\square$

By Lemma 8 we may drop the condition $(k, N) = 1$ from (17) at a cost of $O(N^{o(1)})$. This is essential: expanding $\mathbf{1}_{(k,N)=1}$ by Möbius over $e | N$ would multiply the number of Bombieri–Vinogradov calls by $2^{\omega(N)} = N^{o(1)}$ and destroy the budget.

3.5 Step 4: unfolding μ^2 and the coprimality

Insert $\mu^2(k) = \sum_{d^2|k} \mu(d)$ and $\mathbf{1}_{(k,m)=1} = \sum_{e|(k,m)} \mu(e)$ and truncate at D_0, E_0 of (11).

Two indices are truncated at once, so the discarded set has to be named before it is bounded, and it is not the union of two independent tails. Write the double sum over (d, e) and split it as

$$\sum_{d,e} = \sum_{d \leq D_0} \sum_{e \leq E_0} + \underbrace{\sum_{d > D_0} \sum_e}_{e\text{-sum complete}} + \underbrace{\sum_{d \leq D_0} \sum_{e > E_0}}_{d\text{-sum truncated}}, \quad (18)$$

an exact identity. In the second block the inner e -sum is complete, so it collapses to $\mathbf{1}_{(k,m)=1}$. In the third block the d -sum is truncated and must be bounded as such; no cancellation from the complete identity for $\mu^2(k)$ is used below.

Tail $d > D_0$. Such terms have $d^2 \mid k$ and $mk < N$, so necessarily $d \leq \sqrt{k} < \sqrt{N/m}$. Also $n = N - mk \equiv N \pmod{md^2}$; the number of $n \leq N$ in that progression is $\ll N/(md^2) + 1$ and each $\Lambda \ll \log N$. Hence

$$\ll \log N \sum_{m < M} \sum_{D_0 < d \leq \sqrt{N/m}} \left(\frac{N}{md^2} + 1 \right) \ll \frac{N(\log N)^2}{D_0} + \sqrt{NM} \log N,$$

which is

$$\ll N(\log N)^2 e^{-\sqrt{\log N}} + N^{1-\theta'/2} \log N.$$

Tail $e > E_0$. Here m is squarefree, hence every $e \mid m$ is squarefree. Taking absolute values before summing over the truncated d -range gives moduli $m \operatorname{lcm}(d^2, e)$. Since

$$\frac{1}{\operatorname{lcm}(d^2, e)} = \frac{(d, e)}{d^2 e} \quad (e \text{ squarefree}), \quad \sum_{d \geq 1} \frac{(d, e)}{d^2} \ll \prod_{p \mid e} \left(1 + \frac{1}{p} \right) \ll \log \log(3e),$$

the progression-counting part is

$$\ll N \log N \sum_{m < M} \frac{1}{m} \sum_{\substack{e \mid m \\ e > E_0}} \frac{\log \log(3e)}{e} \ll \frac{N(\log N)^3 \log \log N}{E_0}.$$

The “+1” in the progression count contributes at most

$$\ll D_0 \log N \sum_{m < M} \tau(m) \ll D_0 M (\log N)^2.$$

Thus this tail is $\ll N(\log N)^3 \log \log N e^{-\sqrt{\log N}} + M e^{\sqrt{\log N}} (\log N)^2$.

Both are $\ll N e^{-c\sqrt{\log N}}$ for any $c < 1$, hence below every power of $\log N$. In the remaining range put $L := \operatorname{lcm}(d^2, e)$ and $q := mL$; the conditions $d^2 \mid k$, $e \mid k$ and $n = N - mk$ give $n \equiv N \pmod{q}$, and $mk < N$, $mk \geq \max(N - t, mK)$ give $1 \leq n \leq \mathcal{T}_m(t)$ where

$$\mathcal{T}_m(t) := \min(t, N - mK). \tag{19}$$

Therefore, with $\psi(y; q, a) = \sum_{n \leq y, n \equiv a \pmod{q}} \Lambda(n)$,

$$R(t) = \sum_{m < M} \mu(m) \sum_{d \leq D_0} \mu(d) \sum_{\substack{e \mid m \\ e \leq E_0}} \mu(e) \psi(\mathcal{T}_m(t); q, N) + O\left(N e^{-c\sqrt{\log N}}\right). \tag{20}$$

The error here is the two truncation tails just bounded, and it is exponential: Bombieri–Vinogradov has not been applied yet, and the second sentence of Theorem 1 — that every ingredient but Bombieri–Vinogradov saves exponentially — is read off this line among others. By Lemma 8 we may further restrict to $(q, N) = 1$ at a cost of $O(N^{o(1)})$; the lemma obtains this by counting the admissible n , of which there are $\ll (\log N)^2$, rather than the parameter triples.

Note the size of the modulus: $q = m \operatorname{lcm}(d^2, e) \leq M D_0^2 E_0$, so by (11)

$$q \leq M e^{3\sqrt{\log N}} \ll N^{1/2-2\delta} e^{3\sqrt{\log N}} \leq N^{1/2-\delta} \tag{21}$$

for N large. This is the level at which Bombieri–Vinogradov is applied.

3.6 Step 5: Bombieri–Vinogradov

Lemma 9 (τ -weighted Bombieri–Vinogradov). *For all $A, B > 0$ there is $C = C(A, B)$ such that, for $Q \leq N^{1/2}(\log N)^{-C}$,*

$$\sum_{q \leq Q} \tau_3(q)^B \max_{y \leq N} \max_{(a, q)=1} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \ll_{A, B} \frac{N}{(\log N)^A}.$$

Proof. This is a standard weighted consequence of the maximal Bombieri–Vinogradov theorem; see, for example, [6, §20.2]. Put

$$\mathcal{E}_q := \max_{y \leq N} \max_{(a, q)=1} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right|.$$

By Cauchy–Schwarz,

$$\sum_{q \leq Q} \tau_3(q)^B \mathcal{E}_q \leq \left(\sum_{q \leq Q} \tau_3(q)^{2B} \mathcal{E}_q \right)^{1/2} \left(\sum_{q \leq Q} \mathcal{E}_q \right)^{1/2}.$$

The trivial estimate $\mathcal{E}_q \ll (N/\varphi(q)) \log N$ and the standard mean-value bounds for fixed powers of the divisor function give

$$\sum_{q \leq Q} \tau_3(q)^{2B} \mathcal{E}_q \ll_B N(\log N)^{C_1(B)}.$$

On the other hand, the maximal Bombieri–Vinogradov theorem gives, for every $D > 0$, a constant $C_2(D)$ such that

$$\sum_{q \leq Q} \mathcal{E}_q \ll_D \frac{N}{(\log N)^D} \quad (Q \leq N^{1/2}(\log N)^{-C_2(D)}).$$

Taking $D = 2A + C_1(B) + 2$ and increasing the modulus exponent if necessary proves the stated estimate. No strengthened distribution theorem is being used here; only the usual Bombieri–Vinogradov theorem with an arbitrarily large logarithmic saving. $\square$

A modulus q arises in (20) from at most $\sum_{m|q} \tau(q/m)^2 \leq \tau(q)^3 \leq \tau_3(q)^3$ triples (m, d, e) : given $m \mid q$ there are at most $\tau(q/m)$ choices of d with $d^2 \mid q/m$ and at most $\tau(q/m)$ of e with $\text{lcm}(d^2, e) = q/m$. Writing $\psi(\mathcal{T}_m; q, N) = \mathcal{T}_m/\varphi(q) + \mathcal{E}$ and applying Lemma 9 with $B = 3$ and (21),

$$R(t) = \text{MT}(t) + O\left(\frac{N}{(\log N)^A}\right), \quad \text{MT}(t) = \sum_{\substack{m \leq M \\ (m, N)=1}} \mu(m) \mathcal{T}_m(t) c_{D_0, E_0}(m), \quad (22)$$

where $c_{D_0, E_0}(m) = \sum_{d \leq D_0} \sum_{e|m, e \leq E_0} \mu(d) \mu(e) \mathbf{1}_{(d, N)=1} / \varphi(m \text{lcm}(d^2, e))$.

3.7 Step 6: the density

Lemma 10. *Let m be squarefree with $(m, N) = 1$ and put*

$$c(m) := \sum_{d \geq 1} \sum_{e|m} \frac{\mu(d) \mu(e) \mathbf{1}_{(d, N)=1}}{\varphi(m \text{lcm}(d^2, e))}.$$

Then $c(m) = \mathfrak{A}(N) \lambda(m)/m$ with $\mathfrak{A}(N)$ as in (3) and $\lambda(m) = \prod_{p|m} (1 - \frac{1}{p(p-1)})^{-1}$. Moreover the two truncations contribute differently:

$$c_{D_0, E_0}(m) = c(m) + O\left(\frac{d(m)}{\varphi(m) D_0}\right) + O\left(\frac{d(m) \log \log M}{\varphi(m) E_0}\right). \quad (23)$$

The divisor factor is retained in both truncation errors; it is harmless only after the subsequent sum over m .

Proof. The sum is multiplicative. If $p \nmid m$ then $p \nmid e$ and the local factor is $1 - 1/\varphi(p^2) = 1 - 1/(p(p-1))$ for $p \nmid N$, and 1 for $p \mid N$. If $p \mid m$ (so $p \nmid N$) the p -part of $m \operatorname{lcm}(d^2, e)$ is $p^{1+\max(2v_p(d), v_p(e))}$, and the four choices $(v_p(d), v_p(e)) \in \{0, 1\}^2$ contribute

$$\frac{1}{p-1} - \frac{1}{p(p-1)} - \frac{1}{p^2(p-1)} + \frac{1}{p^2(p-1)} = \frac{1}{p}.$$

Hence $c(m) = \prod_{p \mid m} p^{-1} \cdot \prod_{p \nmid m, p \nmid N} (1 - \frac{1}{p(p-1)}) = m^{-1} \mathfrak{A}(N) \lambda(m)$.

The two truncations are done separately. For $d > D_0$ we retain the complete $e \mid m$ sum only in absolute value. Since $m \operatorname{lcm}(d^2, e)$ is a multiple of md^2 and there are at most $d(m)$ choices of e ,

$$\sum_{d > D_0} \sum_{e \mid m} \frac{1}{\varphi(m \operatorname{lcm}(d^2, e))} \ll \frac{d(m)}{\varphi(m)} \sum_{d > D_0} \frac{1}{d\varphi(d)} \ll \frac{d(m)}{\varphi(m) D_0}.$$

For $e > E_0$, sum first over d . A prime-by-prime comparison with $\varphi(me)$ gives

$$\sum_{d \geq 1} \frac{1}{\varphi(m \operatorname{lcm}(d^2, e))} \ll \frac{\log \log(3e)}{\varphi(me)} = \frac{\log \log(3e)}{e\varphi(m)},$$

since m is squarefree and $e \mid m$, so $\varphi(me) = e\varphi(m)$. Hence

$$\sum_{\substack{e \mid m \\ e > E_0}} \sum_{d \geq 1} \frac{1}{\varphi(m \operatorname{lcm}(d^2, e))} \ll \frac{d(m) \log \log M}{\varphi(m) E_0}. \quad (24)$$

Finally $\sum_{m < M} d(m)/\varphi(m) \ll (\log M)^2$, while $D_0^{-1} = E_0^{-1} = e^{-\sqrt{\log N}}$, so both errors are exponentially small after the m -sum in Proposition 12. $\square$

The load-bearing local factor can equivalently be written, for squarefree m , as

$$\sum_{g \mid m} \frac{\mu(g)}{\varphi(m/g) g \varphi(g)} = \frac{1}{m}.$$

Thus the Euler factor is exactly p^{-1} and the $1/\zeta$ factor appearing in Lemma 11 has first order, which is the cancellation used below.

Lemma 11. *Let $f(m) := \mu(m) \lambda(m) \mathbf{1}_{(m, N)=1}$ and $G(x) := \sum_{m \leq x} f(m)/m$. Then*

$$G(x) \ll \frac{N}{\varphi(N)} e^{-c\sqrt{\log x}} + x^{-1/4} e^{C\sqrt{\log N}} \quad (x \geq 2).$$

Proof. Write $F(s) = \sum_m f(m) m^{-s} = \prod_{p \nmid N} (1 - \lambda(p) p^{-s})$ and $F(s) = \zeta(s)^{-1} H(s)$, so $f = \mu * h$ with h multiplicative, $h(p^j) = 1$ for $p \mid N$ and $h(p^j) = 1 - \lambda(p) = O(p^{-2})$ for $p \nmid N$. Then $G(x) = \sum_{b \leq x} \frac{h(b)}{b} \sum_{a \leq x/b} \frac{\mu(a)}{a}$. For $b \leq \sqrt{x}$ use $\sum_{a \leq y} \mu(a)/a \ll e^{-c\sqrt{\log y}}$, which is the classical zero-free region estimate [4], together with $\sum_b |h(b)|/b \ll \prod_{p \mid N} (1 - 1/p)^{-1} \ll N/\varphi(N)$. For $b > \sqrt{x}$ bound $\sum_{b > y} |h(b)|/b \leq y^{-1/2} \sum_b |h(b)| b^{-1/2} \ll y^{-1/2} \prod_{p \mid N} (1 - p^{-1/2})^{-1} \ll y^{-1/2} e^{C\sqrt{\log N}}$. $\square$

Since $M \asymp N^{1-\theta'}$ is a fixed power of N , the second term of Lemma 11 is negligible at $x \asymp M$, and $N/\varphi(N) \ll \log \log N$; so $G(x) \ll e^{-c'\sqrt{\log x}}$ for x of that size. It is not so for small x : at $x = 2$ the second term is $2^{-1/4} e^{C\sqrt{\log N}}$, and the bound $G(x) \ll e^{-c'\sqrt{\log x}}$ fails on the whole of $x \leq e^{4C\sqrt{\log N}}$. Proposition 12 integrates G over $[1, M]$ and therefore splits there; the split is carried out in its proof.

3.8 Step 7: the main term dies, uniformly in t

Proposition 12. $\sup_{1 \leq t < N} |\text{MT}(t)| \ll N e^{-c\sqrt{\log N}}.$

Proof. By Lemma 10, write

$$\text{MT}(t) = \mathfrak{A}(N) \sum_{m < M} \frac{f(m)}{m} \mathcal{T}_m(t) + \mathcal{E}_{\text{tr}}(t),$$

with $\mathcal{T}_m(t)$ as in (19). The truncation term $\mathcal{E}_{\text{tr}}(t)$ is estimated from the two errors in (23) below. Put $m_0 := (N - t)/K$ and $m_* := \max(1, m_0)$. On the interval $[1, M]$, the function $x \mapsto \mathcal{T}_x(t)$ is constant where $x < m_0$, has derivative $-K$ where $x > m_0$, and satisfies $\mathcal{T}_M(t) = 0$. Since the summation starts at $m = 1$, Abel summation against G of Lemma 11 gives

$$\sum_{m < M} \frac{f(m)}{m} \mathcal{T}_m(t) = G(M) \mathcal{T}_M(t) + K \int_{m_*}^M G(x) dx = K \int_{m_*}^M G(x) dx.$$

Hence, uniformly in t ,

$$\left| \sum_{m < M} \frac{f(m)}{m} \mathcal{T}_m(t) \right| \leq K \int_1^M |G(x)| dx.$$

Lemma 11's second term is at least 1 below $x_0 := e^{4C\sqrt{\log N}}$, so the integral is split there. On $[1, x_0]$ use the trivial $|G(x)| \ll \log x$, giving $\ll K x_0 \log x_0 \ll N^{\theta'} e^{O(\sqrt{\log N})}$. On $[x_0, M]$ the two terms of Lemma 11 give

$$K \int_{x_0}^M \left(\frac{N}{\varphi(N)} e^{-c\sqrt{\log x}} + x^{-1/4} e^{C\sqrt{\log N}} \right) dx \ll K M e^{-c'\sqrt{\log M}} + K M^{3/4} e^{C\sqrt{\log N}},$$

and with $KM = N$, $\log M \asymp \log N$ and $M \asymp N^{1-\theta'}$ the two are $\ll N e^{-c''\sqrt{\log N}}$ and $\ll N^{(3+\theta')/4} e^{C\sqrt{\log N}}$ respectively. All three pieces are $\ll N e^{-c\sqrt{\log N}}$ for $\theta' < 1$, the first and the third being powers of N below N itself. The truncation term introduced above has two halves. Using (23) and $\mathcal{T}_m(t) \leq N$ in both gives

$$|\mathcal{E}_{\text{tr}}(t)| \ll \frac{N}{D_0} \sum_{m < M} \frac{d(m)}{\varphi(m)} + \frac{N \log \log M}{E_0} \sum_{m < M} \frac{d(m)}{\varphi(m)} \ll N (\log M)^2 (1 + \log \log M) e^{-\sqrt{\log N}}.$$

Thus $\mathcal{E}_{\text{tr}}(t) \ll N e^{-c\sqrt{\log N}}$ for every fixed $c < 1$. The size here is N and not M : the truncation is per term of the harmonic sum, and the terms are of the size of N . $\square$

The cancellation is genuine and global: the term $m = 1$ alone contributes $\mathcal{T}_1(t) \asymp N$ — the endpoint of (19), not the weighted sum of Theorem 1 — so the smallness of MT is entirely due to the cancellation in $\sum f(m)/m$.

3.9 Conclusion

Combining (12), (13), (16), (22) and Proposition 12:

$$T_1(t) = \underbrace{P(t)}_{N^{o(1)}} - \underbrace{\text{MT}(t)}_{\ll N e^{-c\sqrt{\log N}}} - \underbrace{O(N(\log N)^{-A})}_{\text{BV}} - \underbrace{C(t)B(K)}_{\ll N e^{-c\sqrt{\log N}}},$$

uniformly in $t < N$. This proves Theorem 1. Every displayed remainder above is $O(N e^{-c\sqrt{\log N}})$ except the term supplied by Lemma 9, which gives the stated arbitrary log-power saving. $\square$

4 Proof of Corollary 2

Fix k and set $a_n := \Lambda(n)\mu(N-n)(\mathbf{1}_{n \equiv N(k)} - 1/\varphi(k))$, so that $E_\mu(t; k) = \sum_{n \leq t} a_n$ and the bracket in (5) is $\sum_{n < N} a_n \log(N-n)$. Since $\log(N-n)$ vanishes at $n = N-1$ and has derivative $-1/(N-t)$, Abel summation gives

$$\sum_{n < N} a_n \log(N-n) = \int_1^{N-1} \frac{E_\mu(t; k)}{N-t} dt.$$

The weight $\log(N-n)$ does not depend on k , so multiplying by $\mu(k)$ and summing over $k < K$ with $(k, N) = 1$ (a finite sum) gives the exact identity

$$E_4(\alpha) = \int_1^{N-1} \frac{T_1(t)}{N-t} dt.$$

Hence $|E_4(\alpha)| \leq (\sup_{t < N} |T_1(t)|) \log N \ll_A N(\log N)^{1-A}$ by Theorem 1. As A is arbitrary this is Corollary 2. Thus, in this regime, the invocation of Huang–Li’s Lemma 5 for E_4 may be replaced by Theorem 1; after the cutoff correction of Section 7, the remaining use of EH_μ within the E_3/E_4 decomposition considered here is the estimate for $E_3(\alpha)$. $\square$

5 Proof of Theorem 3

Take the weight $w_k = \log k$, i.e. the functional E_3 of (4). Steps 0–1 are unchanged, and the complete divisor sum of Lemma 7 is replaced by the following.

Lemma 13. *For every $u \geq 1$, write $u = u_N u'$ where u_N is the largest divisor of u composed of primes dividing N . Then $\sum_{k|u, (k, N)=1} \mu(k) \log k = -\Lambda(u')$.*

Proof. The sum is over $k | u'$ and equals $-\Lambda(u')$ by $\mu * \log = \Lambda$. $\square$

Hence the complete piece of E_3 is

$$- \sum_{1 \leq u < N} \Lambda(N-u) \mu(u) \Lambda(u') = - \sum_{u < N, (u, N)=1} \Lambda(N-u) \mu(u) \Lambda(u) + O(N^{o(1)} \log^2 N),$$

the discarded terms being those with $(u, N) > 1$, which are degenerate in the sense of Lemma 8: a prime $p | (u, N)$ divides $N-u$ as well, so $\Lambda(N-u) \neq 0$ forces $N-u$ to be a power of p , and the number of such u is $\ll (\log N)^2$ while each term is $\ll (\log N)^2$. The set discarded here looks like a set of size N and is not one; that it is not is the same mechanism as Lemma 8, applied to u instead of to k . Now $\mu(u)\Lambda(u)$ is supported on primes $u = p$, where it equals $-\log p$; prime powers $u = p^\ell$, $\ell \geq 2$, have $\mu(u) = 0$. So the complete piece equals

$$\sum_{p < N} \Lambda(N-p) \log p + O(N^{o(1)} \log^2 N) = \sum_{n < N} \Lambda(n) \Lambda(N-n) + O(N^{1/2} \log^2 N), \quad (25)$$

the von–Mangoldt Goldbach convolution, up to the displayed proper-prime-power error.

Two further terms have to be evaluated, and neither vanishes.

(i) *The residual main term.* Steps 3–6 are repeated with the extra factor $\log k$, and what has to be checked is that the factor passes through each of them. It does, for one reason: on the residual range $k = (N-n)/m$, so that

$$\log k = w_m(n), \quad w_m(n) := \log \frac{N-n}{m}, \quad (26)$$

a function of n and m alone. It does not involve d , e , or the modulus $q = m \operatorname{lcm}(d^2, e)$ built from them.

Steps 3 and 4 are therefore unchanged. The factorisation $u = mk$, the degeneracy lemma, and the unfolding of $\mu^2(k)$ and $\mathbf{1}_{(k,m)=1}$ all act on k through its divisors and not through its size, and w_m rides along as a coefficient. Two bookkeeping items move, each by one power of $\log N$: in Lemma 8 a term is now $\ll (\log N)^2$ rather than $\ll \log N$, which leaves that lemma's $O(N^{o(1)})$ where it was, and in the two truncation tails of Step 4 the factor $\log N$ standing for $\max \Lambda$ becomes $(\log N)^2$, so those tails read $\ll N(\log N)^3 e^{-\sqrt{\log N}} + N^{1-\theta'/2}(\log N)^2$ and

$$\ll N(\log N)^4 \log \log N e^{-\sqrt{\log N}} + M e^{\sqrt{\log N}} (\log N)^3,$$

still $\ll N e^{-c\sqrt{\log N}}$ for every fixed $c < 1$. The moduli are the same moduli, so (21) is untouched.

Step 5 is where the factor costs something, and it costs exactly one $\log N$. If $\mathcal{T}_m(t) < 1$, the original inner sum is empty; such an endpoint case can occur for at most one integer m , and its omitted main-term contribution is $O(N^{o(1)})$. We therefore restrict here to $\mathcal{T}_m(t) \geq 1$. Fix m and write $\psi_w(y; q, a) = \sum_{n \leq y, n \equiv a(q)} \Lambda(n) w_m(n)$. On $1 \leq n \leq \mathcal{T}_m(t)$ one has $N - n \geq mK$, so $w_m \geq \log K > 0$, and w_m decreases with $-w'_m(y) = 1/(N - y) \geq 0$; Abel summation gives

$$\psi_w(\mathcal{T}_m; q, N) = w_m(\mathcal{T}_m) \psi(\mathcal{T}_m; q, N) + \int_1^{\mathcal{T}_m} \frac{\psi(y; q, N)}{N - y} dy.$$

Replacing $\psi(y; q, N)$ by $y/\varphi(q)$ in both places costs at most

$$\left(w_m(\mathcal{T}_m) + \int_1^{\mathcal{T}_m} \frac{dy}{N - y} \right) \max_{y \leq N} \left| \psi(y; q, N) - \frac{y}{\varphi(q)} \right| = w_m(1) \mathcal{E}_q \leq (\log N) \mathcal{E}_q,$$

the bracket telescoping to $w_m(1) = \log((N - 1)/m)$. Summing over the triples (m, d, e) exactly as in Step 5 and applying Lemma 9 with $B = 3$ and with $A + 1$ in place of A turns that into $O(N(\log N)^{-A})$. The one extra $\log N$ is absorbed by the arbitrariness of A , which is why no exponent moves.

What is left in place of $\mathcal{T}_m/\varphi(q)$ is $\varphi(q)^{-1}(w_m(\mathcal{T}_m)\mathcal{T}_m + \int_1^{\mathcal{T}_m} y(N - y)^{-1} dy) = \varphi(q)^{-1}(\int_1^{\mathcal{T}_m} w_m(y) dy + w_m(1))$, again by parts; the $w_m(1)$ is $\ll \log N$ per class and, over the $\ll MD_0 E_0 N^{o(1)}$ triples, is $\ll N^{1-\theta'+o(1)}$. Step 6 is unchanged, the density $c(m)$ of Lemma 10 being a property of the modulus and not of the weight. Here the weighted integral is $\ll N \log N$, so the two truncation errors in (23) contribute together

$$\ll N \log N \left(\frac{1}{D_0} \sum_{m < M} \frac{d(m)}{\varphi(m)} + \frac{\log \log M}{E_0} \sum_{m < M} \frac{d(m)}{\varphi(m)} \right) \ll N(\log N)^3 (1 + \log \log M) e^{-\sqrt{\log N}},$$

which is $\ll N e^{-c\sqrt{\log N}}$ for every fixed $c < 1$. Substituting $v = N - y$ in $\int_1^{\mathcal{T}_m} w_m(y) dy$ and taking the actual endpoint $t = N - 1$, for which $\mathcal{T}_m = N - mK$ (since $mK \geq K > 1$ for large N), the outcome is the main term

$$\mathfrak{A}(N) \sum_{m < M} \frac{\mu(m)\lambda(m)}{m} \int_{mK}^N \log(v/m) dv.$$

Substituting $v = mt$ evaluates the integral as $N \log(N/m) - N - mK \log K + mK$. The cancellation estimates must be used here with the same small- x qualification as in Proposition 12. Since $M \asymp N^{1-\theta'}$ is a fixed power of N , Lemma 11 gives $G(M) \ll e^{-c\sqrt{\log N}}$, so the $N \log N$ and $-N$ pieces are $O(N e^{-c'\sqrt{\log N}})$. For $S(M) := \sum_{m \leq M} f(m)$, partial summation gives

$$S(M) = MG(M) - \int_1^M G(x) dx.$$

Split the integral at $x_0 = e^{4C\sqrt{\log N}}$ as in the proof of Proposition 12: use the trivial $G(x) \ll \log x$ below x_0 and Lemma 11 above it. The same estimates used there give $KS(M) \ll Ne^{-c'\sqrt{\log N}}$, which disposes of the $-mK \log K$ and mK pieces. Finally, with $\tilde{G}(M) := \sum_{m \leq M} f(m) \log m/m$ and $\tilde{G}(1)$ denoting the value at $s = 1$ of the differentiated Dirichlet series below, summation by parts on $[M, \infty)$ gives

$$|\tilde{G}(1) - \tilde{G}(M)| \leq |G(M)| \log M + \int_M^\infty \frac{|G(x)|}{x} dx \ll e^{-c'\sqrt{\log N}},$$

because both terms of Lemma 11 are integrable on this tail. Thus only the $-N \log m$ piece contributes to the limiting constant, and what remains is $-\mathfrak{A}(N) N \tilde{G}(1)$ with $\tilde{G}(1) = \lim_x \sum_{m \leq x} \mu(m) \lambda(m) \mathbf{1}_{(m, N)=1} \log m/m$. The constant is fixed by

$$\mathfrak{A}(N) \tilde{G}(1) = -\mathfrak{S}(N), \quad (27)$$

so that the residual main term is $+\mathfrak{S}(N) N + O(N(\log N)^{-A})$.

The sign in (27) is forced, and it is the whole identity: with $F(s) = \sum_m f(m) m^{-s} = \zeta(s)^{-1} H(s)$ as in Lemma 11 one has $\tilde{G}(1) = -F'(1) = -H(1) < 0$, while $\mathfrak{A}(N) H(1) = \mathfrak{S}(N)$ identically — the local factors pair as

$$\left(1 - \frac{1}{p(p-1)}\right) \left(1 - \frac{1}{(p^2-p-1)(p-1)}\right) = 1 - \frac{1}{(p-1)^2}$$

at $p \nmid N$, and as $p/(p-1)$ at $p \mid N$.

(ii) *The subtracted mean term.* It carries the factor

$$B_{\log}(K) := \sum_{\substack{k \leq K \\ (k, N)=1}} \frac{\mu(k) \log k}{\varphi(k)} = -\mathfrak{S}(N) + O(e^{-c\sqrt{\log K}} \log K),$$

which follows from Huang–Li’s Lemma 1: subtracting $\sum_{k \leq K, (k, N)=1} \mu(k) \varphi(k)^{-1} \log(K/k) = \mathfrak{S}(N) + O(e^{-c_1\sqrt{\log K}})$ from $\log K \cdot \sum_{k \leq K, (k, N)=1} \mu(k)/\varphi(k) = O(\log K e^{-c_1\sqrt{\log K}})$ gives the claim. If K is an integer, replacing $k \leq K$ by the defining range $k < K$ changes this by one term of size $K^{-1+o(1)} \log K$, which is absorbed by the same error. The coprimality condition is carried throughout: without it the limiting constant is not $\mathfrak{S}(N)$. Independently: with $f(s) = \prod_{p \nmid N} (1 - \frac{p^{-s}}{p-1})$ one has $f(s) = \zeta(s+1)^{-1} h(s)$ with $h(0) = \mathfrak{S}(N)$, so $B_{\log}(\infty) = -f'(0) = -\mathfrak{S}(N)$. Since the mean term is $-C(N) B_{\log}(K)$ with $C(N) = \sum_{n < N} \Lambda(n) \mu(N-n)$, it contributes $+\mathfrak{S}(N) C(N)$.

Collecting (25), (i) and (ii):

$$E_3(\alpha) = \sum_{n < N} \Lambda(n) \Lambda(N-n) - \mathfrak{S}(N) N + \mathfrak{S}(N) C(N) + O_A\left(\frac{N}{(\log N)^A}\right),$$

which proves Theorem 3. Rearranging, the bound $E_3(\alpha) \ll_A N(\log N)^{-A}$ is equivalent, at this error scale, to

$$\tilde{r}(N) = \mathfrak{S}(N)(N - C(N)) + O_A(N(\log N)^{-A}),$$

which is Huang–Li’s equation (22). Their lower bound (2) is a subsequent consequence of that equation. For the positivity step one needs the quantitative gap, not merely $\mathfrak{A}(N) < 1$. The Bombieri–Vinogradov estimate used by Huang–Li and recalled in the proof of Proposition 5 gives $|C(N)| \leq U(N) = \mathfrak{A}(N) N + O_B(N(\log N)^{-B})$, while the least prime $q_0 \nmid N$ satisfies $q_0 \ll \log N$ and hence $1 - \mathfrak{A}(N) \gg (\log N)^{-2}$. Choosing the error exponents in (22) and in $U(N)$ larger than 2 therefore gives the quantitative lower bound

$$\tilde{r}(N) \gg \frac{N}{(\log N)^2}.$$

On the other hand,

$$\tilde{r}(N) = \sum_{\substack{p+q=N \\ p, q \text{ prime}}} \log p \log q + O(N^{1/2} \log^2 N),$$

because the omitted terms have at least one proper prime power. The error is $o(N(\log N)^{-2})$, so the prime–prime weighted sum is positive for all sufficiently large even N . This is the binary Goldbach implication used here. Moreover, since $\mathfrak{S}(N) \geq 2 \prod_{p>2} (1 - (p-1)^{-2}) \gg 1$, equation (22) gives $\tilde{r}(N) \sim \mathfrak{S}(N)N$ if and only if $C(N) = o(N)$; that latter condition is not proved here. $\square$

6 Two weaker sufficient conditions

Proof of Proposition 5. From Theorem 3,

$$\tilde{r}(N) = \mathfrak{S}(N)(N - C(N)) + E_3(\alpha) + O_A(N(\log N)^{-A}).$$

By the triangle inequality, $|C(N)| \leq U(N) := \sum_{n < N} \Lambda(n) \mu^2(N - n)$, and the Bombieri–Vinogradov calculation in [2, §3.3] gives, for every $B > 0$,

$$U(N) = \mathfrak{A}(N)N + O_B(N(\log N)^{-B}).$$

Therefore

$$\tilde{r}(N) \geq \mathfrak{S}(N)(1 - \mathfrak{A}(N))N + E_3(\alpha) + O_B(\mathfrak{S}(N)N(\log N)^{-B}) + O_A(N(\log N)^{-A}).$$

The least prime $q_0 \nmid N$ satisfies $q_0 \ll \log N$, hence

$$1 - \mathfrak{A}(N) \geq \frac{1}{q_0(q_0 - 1)} \gg (\log N)^{-2}, \quad \mathfrak{S}(N) \geq 2C_2 > 0.$$

Substituting (8) and taking $A, B > 2$ leaves a positive main term of size at least a fixed multiple of $N(\log N)^{-2}$ for all sufficiently large N . Since the contribution to $\tilde{r}(N)$ from terms with a proper prime power is $O(N^{1/2} \log^2 N) = o(N(\log N)^{-2})$, the prime–prime weighted Goldbach sum is positive. Hence binary Goldbach follows. $\square$

Proof of Proposition 6. Since $|E_3(\alpha)| \leq B_\mu(N)$, the same calculation gives

$$\tilde{r}(N) \geq \mathfrak{S}(N)(1 - \mathfrak{A}(N))N - B_\mu(N) + o(\mathfrak{S}(N)(1 - \mathfrak{A}(N))N).$$

Equation (10) leaves a positive margin $\gg_\varepsilon N(\log N)^{-2}$. As above, this dominates the $O(N^{1/2} \log^2 N)$ proper-prime-power contribution, so the prime–prime weighted Goldbach sum is positive for all sufficiently large even N .

Finally, the full hypothesis $EH_\mu(N^{\theta'})$ gives, for every $L > 0$,

$$B_\mu(N) \leq (\log K) \sum_{\substack{k < K \\ (k, N) = 1}} |E_\mu(N; k)| \ll_L \frac{N}{(\log N)^{L-1}}.$$

Taking $L > 3$ and using $\mathfrak{S}(N)(1 - \mathfrak{A}(N))N \gg N(\log N)^{-2}$ shows that (10) holds eventually, for example with any fixed $0 < \varepsilon < 1$ after increasing N . $\square$

7 The correction Δ to equation (18)

The arXiv versions v1 and v2 of [2] replace an n -dependent cutoff in the derivation of their equation (18) by the fixed outer range $k < (N - 1)/\alpha$. We do not claim discovery of this defect: it had been reported to the authors earlier, and Moya [7] subsequently gave a finite counterexample and a repair that keeps the moving endpoint. The purpose here is narrower. We write the difference between the fixed-cut and moving-cut formulations as an explicit term Δ , bound it, and use the same estimate to transfer Theorem 1 to the corrected cutoff.

Huang–Li define

$$S_2(\alpha) = \sum_{n < N} \Lambda(n) \mu^2(N - n) \tilde{\Lambda}_\alpha(N - n), \quad \tilde{\Lambda}_\alpha(u) = \sum_{d|u, d > \alpha} \mu(d) \log(1/d),$$

and substitute $k = u/d$, so that the constraint $d > \alpha$ becomes

$$k < \frac{N - n}{\alpha}, \tag{28}$$

an n -dependent bound. Their displayed equation (18) reads

$$S_2(\alpha) = \sum_{k < \frac{N-1}{\alpha}} \mu(k) \sum_{\substack{n < N \\ n \equiv N \pmod{k}}} \Lambda(n) \mu(N - n) \log\left(\frac{k}{N - n}\right),$$

in which (28) has been replaced by the n -free bound $k < (N - 1)/\alpha$. The two differ. Writing $N - n = mk$, the condition (28) is $m > \alpha$, so the right-hand side of (18) contains, in addition to $S_2(\alpha)$, exactly the terms with $m \leq \alpha$. On those terms $\mu(k)\mu(N - n) = \mu(m)\mu^2(k)\mathbf{1}_{(k,m)=1}$ and $\log(k/(N - n)) = -\log m$, so

$$S_2(\alpha) = \text{RHS of (18)} + \Delta, \quad \Delta = \sum_{2 \leq m \leq \alpha} \mu(m) \log m \sum_{\substack{k < (N-1)/\alpha \\ (k,m)=1}} \mu^2(k) \Lambda(N - mk). \tag{29}$$

(The term $m = 1$ drops out since $\log 1 = 0$.) The trivial bound is $\Delta \ll N(\log N)^2$, which is far above the target $O(N(\log N)^{-A})$. Thus the trivial estimate does not make Δ negligible, and the term needs its own treatment.

It does close, by the machinery already used above and under hypotheses Huang–Li already assume. Indeed Δ has exactly the shape of the residual (17): the Möbius factor sits on the *short* variable $m \leq \alpha$, and the long variable k carries only $\mu^2 \geq 0$. Since $\mu(m) \neq 0$ on the range of summation, the two conditions on k collapse into one,

$$\mu^2(k) \mathbf{1}_{(k,m)=1} = \mu^2(mk), \tag{30}$$

and expanding $\mu^2(mk) = \sum_{b^2 | mk} \mu(b)$ with the truncation $b \leq B_1 := (\log N)^{A+4}$ — the same one Huang–Li use in their §3.1, where it is called B ; the subscript keeps it apart from the free exponent of Lemma 9 — reduces Δ to sums of Λ over progressions to moduli $[m, b^2] \leq \alpha B_1^2 = \alpha(\log N)^{2A+8}$. The tail $b > B_1$ is discarded as the d -tail of Step 4 was, but counted on the pair (m, k) and not on the modulus alone. Such terms have $b^2 | mk$, which holds exactly when $k \equiv 0$ to the modulus $b^2/(b^2, m)$; so for each m the number of admissible $k < K := (N - 1)/\alpha$ is at most $K(b^2, m)/b^2$, and

$$\sum_{m \leq \alpha} (b^2, m) = \sum_{g|b^2} g \cdot \#\{m \leq \alpha : (b^2, m) = g\} \leq \sum_{g|b^2} g \cdot \frac{\alpha}{g} = \alpha d(b^2).$$

Hence, with $\Lambda \ll \log N$, the factor $\log m \leq \log \alpha \ll \log N$ that Δ carries on the short variable, and $K\alpha = N - 1$,

$$\ll (\log N)^2 \sum_{b > B_1} \frac{K\alpha d(b^2)}{b^2} \ll N(\log N)^2 \sum_{b > B_1} \frac{d(b^2)}{b^2} \ll \frac{N(\log N)^2 (\log B_1)^2}{B_1} \ll \frac{N(\log \log N)^2}{(\log N)^{A+2}},$$

which is $\ll_A N(\log N)^{-A}$. The last step is $\sum_{b > X} d(b^2)b^{-2} \ll (\log X)^2/X$, by partial summation from $\sum_{b \leq Y} d(b^2) \ll Y(\log Y)^2$.

Counting on the pair removes any extra distributional input from this tail estimate. Bounding each progression separately by $N/[m, b^2] + 1$ with $[m, b^2] \geq b^2$ drops the factor $1/m$ and drops the constraint that an admissible k exists at all; with the factors $\Lambda \ll \log N$ and $\log m \ll \log N$, the resulting $+1$ then contributes $\ll \alpha \sqrt{N}(\log N)^2$. That crude estimate would therefore require $\alpha < \sqrt{N}$ up to logarithmic factors; the pair count above avoids this auxiliary restriction. What is left is the progression main term. For fixed m the k -range has length $K = (N - 1)/\alpha$, hence the corresponding $n = N - mk$ interval has length mK . Multiplying this by the local density $\mathfrak{A}(N)\lambda(m)/m$ gives

$$\mathfrak{A}(N)K \sum_{m \leq \alpha} \mu(m)\lambda(m) \log m,$$

up to the already bounded progression errors. Partial summation from Lemma 11 makes this $O(Ne^{-c\sqrt{\log N}})$.

The density here is not the one Lemma 10 proves. That lemma evaluates the first of

$$\sum_{d \geq 1} \sum_{e|m} \frac{\mu(d)\mu(e)\mathbf{1}_{(d,N)=1}}{\varphi(m \operatorname{lcm}(d^2, e))}, \quad \sum_{b \geq 1} \frac{\mu(b)}{\varphi([m, b^2])},$$

and what the expansion above leaves is the second, over the moduli restricted to $(\cdot, N) = 1$. The two agree, and the check is the same local computation: for squarefree m the factor of the second is $\frac{1}{p-1} - \frac{1}{p(p-1)} = \frac{1}{p}$ at $p \mid m$ and $1 - \frac{1}{p(p-1)}$ at $p \nmid m$, and the restriction sets the latter to 1 at $p \mid N$ exactly as in Lemma 10 — the same coprimality point used above. The product is $\mathfrak{A}(N)\lambda(m)/m$, as written.

The resulting moduli stay within the level supplied by the corresponding Huang–Li hypothesis. Hence:

- In the Corollary-1 regime, $\alpha = N^{1-\theta'} < N^{1/2}$ and Bombieri–Vinogradov bounds Δ without an additional Elliott–Halberstam hypothesis.
- In the general Theorem-1 regime, $\alpha \asymp N^\theta$ and Δ closes under the hypothesis $EH(N^\theta(\log N)^{2A+8})$, which is assumed there.

Thus the cutoff defect does not by itself invalidate the stated conditional conclusions: Moya repairs the moving-cut argument under the original hypotheses, while the outside formulation above controls the extra term in the same regimes. In the corrected moving-cut formulation Δ does not arise.

7.1 The two formulations differ by Δ

The corrected argument keeps the moving cut: with $Y_k = \lceil N - \alpha k \rceil - 1$ the inner sum stops at Y_k , the terms with $m \leq \alpha$ never enter, and (29) has nothing to correct. Every statement above was proved against the fixed cut, so one must check separately what survives the move. In the Corollary-1 regime the following proposition gives that transfer, and the term carrying the difference has the same short-variable shape as Δ .

Write $J = T_1(N - 1)$ for the fixed-cut object at its extreme endpoint and $T_1^* = \sum_{k < K, (k, N)=1} \mu(k) E_\mu(Y_k; k)$ for the moving-cut one. The difference is carried by the n the moving cut drops, and there

$$n > Y_k \iff N - n \leq \alpha k \iff m \leq \alpha, \quad m = \frac{N - n}{k},$$

so on exactly those terms $\mu(k)\mu(N - n) = \mu(k)\mu(mk)$, which vanishes unless mk is squarefree and equals $\mu(m)\mu^2(k)$ otherwise: the Möbius factor on the *short* variable, the long one carrying only $\mu^2 \geq 0$. That is (29)'s shape with weight 1 in place of $\log m$.

Proposition 14 (the bound survives the corrected cut). *Fix $\theta' \in (1/2, 1)$, put $\alpha = N^{1-\theta'}$ and $K = (N - 1)/\alpha$. Then for every $A > 0$, $T_1^* \ll_{A, \theta'} N(\log N)^{-A}$, unconditionally.*

Proof. Exchanging as above, $J - T_1^* = \Xi_{\text{main}} - \Xi_{\text{mean}}$ with

$$\Xi_{\text{main}} = \sum_{m \leq \alpha} \mu(m) \sum_{\substack{k < K \\ (k, mN)=1}} \mu^2(k) \Lambda(N - mk), \quad \Xi_{\text{mean}} = \sum_{\substack{k < K \\ (k, N)=1}} \frac{\mu(k)}{\varphi(k)} (C(N - 1) - C(Y_k)),$$

the constraint $n \geq 1$ being automatic since $m \leq \alpha$ and $k < K$ force $mk < N - 1$.

$J \ll_{A, \theta'} N(\log N)^{-A}$ is Theorem 1 at $t = N - 1$, which lies in the range of its supremum, and it is used exactly where Theorem 1 needs it. It is one of the two places this proof needs $\theta' > 1/2$; the other is the bridge term below, whose moduli carry the same inequality.

In Ξ_{main} the term $m = 1$ is $\sum_{k < K, (k, N)=1} \mu^2(k) \Lambda(N - k) \leq \sum_{N - K < j < N} \Lambda(j) \ll K \log N = N^{\theta'} \log N$, which is $\ll_A N(\log N)^{-A}$ because $\theta' < 1$. The terms $2 \leq m \leq \alpha$ are (29) with $\log m$ replaced by 1 and close by the argument above, the main term dying by Lemma 11 applied directly — the weight being 1, no partial summation is needed to remove a $\log m$. One step is not verbatim: there the inner variable carries $(k, m) = 1$ and (30) absorbs it into $\mu^2(mk)$, while here it carries $(k, mN) = 1$, and the $(k, N) = 1$ that is left over after (30) is discarded by Lemma 8 at a cost of $O(N^{o(1)})$, as at (17).

For Ξ_{mean} , put $K_- := \lceil K \rceil - 1$, so that for integer k the condition $k < K$ is exactly $k \leq K_-$. Since $n > Y_k$ iff $k > R_n := \lceil (N - n)/\alpha \rceil - 1$, exchanging gives $\Xi_{\text{mean}} = \sum_{n < N} a_n (\rho_N(K_-) - \rho_N(R_n))$, where

$$\rho_N(R) := \sum_{\substack{d \leq R \\ (d, N)=1}} \frac{\mu(d)}{\varphi(d)} \ll e^{-c_1 \sqrt{\log R}} \quad (\log N \ll \log R)$$

by [2, Lemma 1], a form of [1, Lemma 2.1] — the hypothesis $\log N \ll \log R$ is part of the statement and is checked below — and $a_n = \Lambda(n)\mu(N - n)$. Split at $N - n = H$ with $H := 2N^{1-\theta'/2}$. On $N - n \leq H$ use the elementary bound $|\rho_N(R)| \leq \sum_{d \leq R} 1/\varphi(d) \ll \log(2 + R) \ll \log N$ and $\sum_{N-H \leq n < N} |a_n| \ll H \log N$. This contributes $\ll H(\log N)^2 \ll_A N(\log N)^{-A}$ since $1 - \theta'/2 < 1$. On $N - n > H$ one has $R_n \geq N^{\theta'/2}$, so both arguments $K_- \asymp K$ and R_n of ρ_N meet its hypothesis against N , and both terms are $\ll e^{-c\sqrt{\theta' \log N}}$; with $\sum_{n < N} |a_n| \ll N$ this is $\ll_A N(\log N)^{-A}$ as well. $\square$

8 Relation to prior work

Murty and Vatswani [3] use the same convolutional mechanism in the shifted-prime setting, and Huang–Li [2] adapt it to the Goldbach shift. Theorem 3 claims no novelty for that chain; it keeps the surviving E_3 term explicit. The fixed-class feature is also present in [3, 2].

More recently, Cantarini [5] studies weighted averages of the diagonal Möbius-twisted Elliott–Halberstam problem, under GRH and additional hypotheses, for Sobolev and Hölder–Zygmund weights. That work concerns weighted diagonal averages over moduli. The present

Theorem 1 instead exploits the particular signed fixed-class combination occurring in Huang–Li’s E_4 term and, in the Corollary-1 regime, obtains its required bound unconditionally from divisor switching and Bombieri–Vinogradov. We make no claim that this mechanism supplies a corresponding bound for the full diagonal EH_μ average studied there.

The moving-cut defect is not claimed as new: Moya [7] records it and repairs the original argument under Huang–Li’s hypotheses. Section 7 gives a complementary outside correction. The net result is therefore limited: the E_4 use of EH_μ is removed in the regime studied, while the surviving E_3 correlation remains unestimated. No unconditional Goldbach conclusion is claimed.

References

- [1] D. Goldston and C. Yıldırım, *Higher correlations of divisor sums related to primes I*, *Integers* **3** (2003), A5.
- [2] J.-J. Huang and H. Li, *On the connection between the Goldbach conjecture and the Elliott–Halberstam conjecture*, arXiv:2005.03811 [math.NT]; v1 (May 2020), v2 (August 2022). Also published in a Springer volume, doi:10.1007/978-3-030-67996-5_17; references here are to the arXiv version.
- [3] M. R. Murty and A. Vatwani, *Twin primes and the parity problem*, *J. Number Theory* **180** (2017), 643–659.
- [4] H. L. Montgomery and R. C. Vaughan, *Multiplicative number theory I: classical theory*, Cambridge Studies in Advanced Mathematics **97**, Cambridge University Press, 2007.
- [5] M. Cantarini, *Averages of diagonal Elliott–Halberstam problem twisted by Möbius function with Sobolev and Hölder–Zygmund weights*, arXiv:2607.09110 [math.NT], July 2026.
- [6] H. L. Montgomery and R. C. Vaughan, *Multiplicative number theory II: primes and sieves*, Cambridge Studies in Advanced Mathematics **218**, Cambridge University Press, 2026. The maximal Bombieri–Vinogradov theorem used here is from §20.2.
- [7] R. Moya, *A Moving-Cut Correction to the Huang–Li Conditional Goldbach Argument: Consequences for Diagonal Möbius-Twisted Elliott–Halberstam Hypotheses*, Zenodo, August 2026; DOI 10.5281/zenodo.21939906. Gives a finite counterexample to the fixed-cut rearrangement and a moving-endpoint repair under Huang–Li’s original hypothesis.

Disclosure statement. No potential competing interest, financial or non-financial, was reported by the author. No funding was received for this work. AI-assisted drafting and computational auditing were used in preparing the manuscript; no computation or AI-generated output is used as a premise of a proof.

Data availability statement. The audit code, stored outputs, and Lean 4 development accompanying this manuscript are openly available at <https://github.com/soke91/arithmetic-convolution-fields>, in the directory P1-mobius-fixed-class/. No data derived from human or animal subjects were used.